Schur's Lemma and Matrix Orthogonality

Week 5 · Lecture 9

Mehmet Özkan

Group Theory

Reading for Lecture 9

- **Zee, II.2, pp. 101–104:** Schur's lemma, averaging and the Great Orthogonality Theorem. See also Appendix 1, pp. 109–110, for inequivalent representations.
- **Bünemann, §4.1.2, pp. 48–50:** the two forms of Schur's lemma.
- **Bünemann, §5.1, pp. 57–59:** matrix-element orthogonality and its proof.

We first explain the statements, then derive them. The alternative Hermitian proof and the real-field example are optional. All references use printed page numbers.

Group Theory

What we need from Week 4

A representation assigns a matrix $D(g)$ to each $g \in G$, with

$$D(gh) = D(g)D(h), \quad D(e) = I, \quad D(g^{-1}) = D(g)^{-1}.$$

It is irreducible when no nonzero proper subspace is preserved by every $D(g)$. Equivalence means one common basis change:

$$D'(g) = S^{-1}D(g)S \qquad \text{for every } g \in G.$$

For finite groups over $\mathbb{C}$, every representation can be made unitary and decomposes into irreducible pieces. Today we use these results to compare the pieces and their matrix entries.

Group Theory

Conventions and indices

Throughout this week, G is finite, $N = |G|$, and all representation spaces are finite dimensional over $\mathbb{C}$ unless stated otherwise.

Choose one representative $D^{(\alpha)}$ from each equivalence class of irreducible representations. Its dimension is d_α.

$$D^{(\alpha)}(g)_{ij}, \qquad i, j = 1, \ldots, d_\alpha.$$

The label α specifies a representation. The indices i, j specify a matrix entry. They do not label group elements.

Sums are explicit. For a scalar z, z^* denotes complex conjugation; $A^\dagger$ denotes the conjugate transpose of a matrix.

Group Theory

Unitary representations: what is being assumed?

We may choose a basis in which

$$D(g)^\dagger D(g) = I, \qquad D(g^{-1}) = D(g)^\dagger.$$

This follows from the finite-group unitarity theorem proved in W4. It is a statement about a suitable common basis, not about every original set of matrices.

The scalar and intertwiner forms of Schur's lemma below do not require unitarity. The form of matrix orthogonality written with complex-conjugated entries does use unitary representatives.

Textbook: Zee, II.1, pp. 96–97; Bünemann, §4.1.1, pp. 46–47.

Group Theory

Invariant subspaces and invariant complements

Recall that W is invariant when $D(g)w \in W$ for all $w \in W$ and all $g \in G$. If D is unitary and $u \in W^{\perp}$, then

$$\langle w, D(g)u \rangle = \langle D(g^{-1})w, u \rangle = 0.$$

The last equality holds since $D(g^{-1})w \in W$. Thus $W^{\perp}$ is invariant too, and a basis adapted to $W \oplus W^{\perp}$ makes every $D(g)$ block diagonal.

For a finite group, we can first choose a unitary basis and then repeat this splitting. This is the complete reducibility result we will use in Lecture 10.

Group Theory

Matrices that commute with the whole representation

Let $A : V \to V$ be a linear map. We ask when

$$AD(g) = D(g)A \qquad \text{for every } g \in G.$$

The identity and every scalar multiple λI satisfy this. Are other choices possible?

Commuting with one matrix is a weaker condition. If G is generated by r, s, however, it is enough to check that A commutes with both $D(r)$ and $D(s)$: the property then passes to every product and inverse.

The matrices commuting with all $D(g)$ form the **commutant**. Schur's lemma determines it for an irreducible complex representation.

Group Theory

Schur's lemma: the scalar form

Let D be an irreducible representation (an **irrep**) on a nonzero finite-dimensional complex space V. If

$$AD(g) = D(g)A \qquad \text{for every } g \in G,$$

then

$$\boxed{A = \lambda I_V \quad \text{for some } \lambda \in \mathbb{C}.}$$

The same scalar acts on every vector in V. The matrix A need not be Hermitian, unitary or invertible. The conclusion includes $A = 0$.

Textbook: Zee, II.2, pp. 101–102; Bünemann, §4.1.2.2, p. 50.

Group Theory

Why an eigenspace is invariant

Because V is a nonzero finite-dimensional complex space, A has at least one eigenvalue λ. Its eigenspace is

$$E_\lambda = \{v \in V : Av = \lambda v\}.$$

For $v \in E_\lambda$, commutation gives

$$A(D(g)v) = D(g)Av = \lambda D(g)v.$$

Therefore $D(g)v \in E_\lambda$ for every g. The eigenspace is a nonzero invariant subspace of the representation.

We have not assumed that A has a complete basis of eigenvectors.

Group Theory

Proof of the scalar form

Irreducibility allows only the invariant subspaces $\{0\}$ and V. Since $E_\lambda \neq \{0\}$, the previous argument forces

$$E_\lambda = V.$$

Thus $Av = \lambda v$ for every $v \in V$, which is exactly $A = \lambda I$.

The proof uses the existence of one eigenvalue and the invariance of its eigenspace. It does not assume diagonalizability of A or simultaneous diagonalizability of the $D(g)$.

Textbook: Compare Bünemann, §4.1.2, pp. 49–50.

Group Theory

Optional: Zee's Hermitian reduction

In a unitary basis, $AD(g) = D(g)A$ implies that $A^\dagger$ also commutes with every $D(g)$. Take the adjoint, use $D(g)^\dagger = D(g^{-1})$, and note that g^{-1} runs over G as g does.

Write

$$A = H + iK, \qquad H = \frac{A + A^\dagger}{2}, \qquad K = \frac{A - A^\dagger}{2i}.$$

Both H and K are Hermitian and commute with every $D(g)$. It therefore suffices to prove the scalar conclusion for a commuting Hermitian matrix.

Textbook: Zee, II.2, p. 101.

Group Theory

Optional: Zee's eigenvalue-block argument

Choose an orthonormal eigenbasis of H, so $H = \mathrm{diag}(h_1, \ldots, h_d)$. The equation $HD(g) = D(g)H$ becomes

$$(h_i - h_j)D(g)_{ij} = 0 \qquad \text{for every } i, j, g.$$

If two eigenvalues differ, all entries connecting their eigenspaces vanish for every g. Grouping equal eigenvalues would therefore split the representation into invariant blocks.

Irreducibility excludes this split. Hence all h_i are equal. Applying the same reasoning to K gives $A = (h + ik)I$.

Textbook: Zee, II.2, pp. 101–102.

Group Theory

Example: the commutant of the plane representation

For our $D_3 \cong S_3$ representation, use

$$R = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The equation $AF = FA$ gives $b = c = 0$. With $A = \mathrm{diag}(a, d)$, the equation $AR = RA$ then gives

$$-\frac{\sqrt{3}}{2}a = -\frac{\sqrt{3}}{2}d \quad \Longrightarrow \quad a = d.$$

Thus $A = aI_2$, in agreement with Schur's lemma. Commuting with the two generators was enough.

Group Theory

Why irreducibility matters

If $D = D^{(\alpha)} \oplus D^{(\beta)}$, then

$$A = \begin{pmatrix} \lambda I_{d_\alpha} & 0 \\ 0 & \mu I_{d_\beta} \end{pmatrix}$$

commutes with every $D(g)$, even when $\lambda \neq \mu$. It acts differently on the two invariant blocks.

Repeated copies allow still more freedom. If $D(g) = D_E(g) \oplus D_E(g)$, then the matrix exchanging the two copies,

$$T = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix},$$

commutes with every $D(g)$ and is not a scalar matrix.

Group Theory

Optional: why the field $\mathbb{C}$ matters

The real rotation representation of $\mathbb{Z}_3$ on $\mathbb{R}^2$ is irreducible: a 120° rotation has no real eigenvector, hence no invariant real line.

Nevertheless,

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad JR = RJ,$$

and J is not a real scalar multiple of I. Its eigenvalues are $\pm i$, so the eigenvalue proof cannot begin over $\mathbb{R}$.

Over $\mathbb{C}$, the rotation representation splits into two invariant eigenlines. The scalar form of Schur's lemma used here applies to complex irreducible representations.

Group Theory

Symmetry and degenerate eigenvalues

Suppose H is Hermitian, $H(W) \subseteq W$, and $H|_W$ commutes with an irreducible symmetry representation on W. Schur's lemma gives

$$H|_W = EI_W, \qquad E \in \mathbb{R}.$$

All vectors in that block have the same eigenvalue. This is the linear-algebra origin of symmetry-enforced degeneracy in many physical applications.

With repeated equivalent copies, H may mix the copies before this restriction is made; see the exchange matrix T on "Why irreducibility matters." First diagonalize its action on the copy labels.

Textbook: Zee, II.2, p. 102; physical applications follow in Part III.

Group Theory

Maps between two representations

Let $D^{(\alpha)}$ act on V_α and $D^{(\beta)}$ on V_β. A linear map $A : V_\beta \to V_\alpha$ is an **intertwiner** if

$$D^{(\alpha)}(g)A = AD^{(\beta)}(g) \qquad \text{for every } g.$$

Thus transforming before or after applying A gives the same result. The matrix A has d_α rows and d_β columns and may be rectangular.

When the two representations are the same, this is the commuting-matrix condition. The general case will let us compare different irreducible representations.

Textbook: Bünemann, §4.1.2.1, pp. 49–50.

Group Theory

The kernel and image of a linear map

For $A : V_\beta \to V_\alpha$, define

$$\ker A = \{v \in V_\beta : Av = 0\}, \qquad \operatorname{im} A = \{Av : v \in V_\beta\}.$$

Both are vector subspaces. A zero kernel means that A is injective. The condition $\operatorname{im} A = V_\alpha$ means that it is surjective.

This is a linear-map kernel, consisting of vectors. The representation kernel $\ker D = \{g : D(g) = I\}$ from W4 consists of group elements. The two uses of “kernel” have different domains.

Group Theory

An intertwiner has invariant kernel and image

If $Av = 0$, then

$$AD^{(\beta)}(g)v = D^{(\alpha)}(g)Av = 0.$$

Thus $\ker A$ is invariant under $D^{(\beta)}$.

If $u = Av$ lies in $\operatorname{im} A$, then

$$D^{(\alpha)}(g)u = D^{(\alpha)}(g)Av = AD^{(\beta)}(g)v \in \operatorname{im} A.$$

Thus $\operatorname{im} A$ is invariant under $D^{(\alpha)}$. These statements use only the intertwining relation and linearity.

Group Theory

Schur's lemma: the intertwiner form

Suppose both representations are irreducible and $A \neq 0$. Their invariant kernel and image must satisfy

$$\ker A = \{0\}, \qquad \operatorname{im} A = V_\alpha.$$

Indeed, $\ker A = V_\beta$ or $\operatorname{im} A = \{0\}$ would imply $A = 0$.

Hence A is both injective and surjective, so $d_\alpha = d_\beta$ and A is invertible. Therefore an intertwiner between irreducible representations is either zero or an isomorphism.

Textbook: Bünemann, §4.1.2.1, pp. 49–50; Zee, II.2, Appendix 1, pp. 109–110.

Group Theory

Intertwiners and equivalent representations

For a nonzero intertwiner between irreducible representations,

$$D^{(\alpha)}(g)A = AD^{(\beta)}(g) \quad \Longrightarrow \quad D^{(\beta)}(g) = A^{-1}D^{(\alpha)}(g)A.$$

Thus the representations are equivalent. In particular,

$$\boxed{\alpha \neq \beta \quad \Longrightarrow \quad A = 0}$$

when our labels denote inequivalent irreducible representatives.

For the same chosen representative, every intertwiner is λI by the scalar form. Equivalent matrices written in different bases need not have scalar intertwiners.

Group Theory

Exercise: a commuting matrix for the permutation action

Let $P(g)$ be the three-dimensional permutation representation of S_3 introduced in W4. Let J_3 denote the 3×3 matrix with every entry equal to 1.

1. Show that $J_3 P(g) = P(g) J_3 = J_3$ for every permutation.
2. Find the eigenvalue of J_3 on the constant-vector line W_0 and on the zero-sum plane W_2.
3. Why does this not contradict Schur's lemma? What is the general form of a matrix commuting with every $P(g)$?

Use $P \simeq 1 \oplus D_E$. The subscripts in W_0, W_2 are W4 labels, not eigenvalues.

Group Theory

Exercise: solution

Permuting the rows or columns of J_3 leaves it unchanged. Also,

$$J_3 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (x+y+z) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Its eigenvalues are 3 on W_0 and 0 on W_2. The permutation representation is reducible, so the scalar conclusion for the whole space does not apply.

The two blocks are irreducible and inequivalent. Schur's lemma forbids maps between them and allows one scalar on each. In the original coordinates the general answer is

$$A = aI_3 + bJ_3, \qquad a, b \in \mathbb{C}.$$

Group Theory

Averaging a matrix over the group

For a unitary representation D and any square matrix X, define

$$\mathcal{A}_D(X) = \frac{1}{N} \sum_{g \in G} D(g) X D(g)^\dagger .$$

Zee uses $A = \sum_g D(g)^\dagger X D(g)$. Relabelling $g \mapsto g^{-1}$ gives $A = N \mathcal{A}_D(X)$; our factor $1/N$ normalizes the sum.

The average commutes with every $D(h)$. For irreducible D, Schur's lemma fixes it up to a scalar, which the trace determines.

Textbook: Zee, II.2, p. 103; Bünemann, §5.1, pp. 58–59.

Group Theory

The rearrangement lemma in a group sum

For fixed $h \in G$, the map $g \mapsto hg$ is a bijection of G: its inverse is $k \mapsto h^{-1}k$. Therefore

$$\sum_{g \in G} f(hg) = \sum_{k \in G} f(k).$$

The terms may appear in a different order, but each group element still appears exactly once. The same is true of right multiplication $g \mapsto gh$.

This is where the group structure enters the averaging argument. We sum over the entire finite group, including all products of the generators.

Group Theory

Why the average commutes with every symmetry

For any $h \in G$, the representation law gives

$$\begin{aligned} D(h)\mathcal{A}_D(X)D(h)^\dagger &= \frac{1}{N}\sum_g D(hg)XD(hg)^\dagger \\ &= \mathcal{A}_D(X). \end{aligned}$$

The last equality is a relabelling of the group sum. Multiplying on the right by $D(h)$ gives

$$D(h)\mathcal{A}_D(X) = \mathcal{A}_D(X)D(h).$$

Thus the average is a commuting matrix. For irreducible D, it must be a scalar multiple of I.

Group Theory

The trace determines the average

Write $\mathcal{A}_D(X) = \lambda I_d$ for an irreducible representation. Cyclicity and unitarity give

$$\operatorname{tr} \mathcal{A}_D(X) = \frac{1}{N} \sum_g \operatorname{tr}(X D(g)^\dagger D(g)) = \operatorname{tr} X.$$

Since $\operatorname{tr}(\lambda I_d) = \lambda d$, we obtain

$$\boxed{\mathcal{A}_D(X) = \frac{\operatorname{tr} X}{d} I_d.}$$

In particular, averaging a traceless matrix gives zero. Averaging a matrix of trace one gives I_d/d.

Textbook: Zee, II.2, p. 103.

Group Theory

Example: averaging a preferred direction in the plane

For D_3, take $X = \mathrm{diag}(1, 0)$. Since $FXF = X$, the six terms reduce to an average over the three rotations:

$$\mathcal{A}_D(X) = \frac{1}{3}(X + RXR^{-1} + R^2XR^{-2}).$$

Direct multiplication gives

$$RXR^{-1} = \begin{pmatrix} 1/4 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 3/4 \end{pmatrix}, \quad R^2XR^{-2} = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}.$$

The off-diagonal entries cancel. Each diagonal average is $1/2$, giving $\mathcal{A}_D(X) = I_2/2$. This agrees with $\mathrm{tr}\, X = 1$ and $d = 2$.

Group Theory

Averaging between different irreducible spaces

For a $d_\alpha \times d_\beta$ matrix X, define

$$\mathcal{A}_{\alpha\beta}(X) = \frac{1}{N} \sum_g D^{(\alpha)}(g) X D^{(\beta)}(g)^\dagger.$$

The same relabelling proves that this matrix intertwines the two representations. Therefore

$$\mathcal{A}_{\alpha\beta}(X) = \begin{cases} 0, & \alpha \neq \beta, \\ (\operatorname{tr} X / d_\alpha) I_{d_\alpha}, & \alpha = \beta. \end{cases}$$

The trace is used only in the second case, when X is square and both matrices use the same chosen representative.

Textbook: Bünemann, §5.1, pp. 58–59.

Group Theory

Selecting a single matrix entry

Let E_{jl} be the rectangular matrix with a single 1 in row j, column l, and zeros elsewhere:

$$(E_{jl})_{ab} = \delta_{aj}\delta_{bl}.$$

Then matrix multiplication selects two entries:

$$\left(D^{(\alpha)}(g)E_{jl}D^{(\beta)}(g)^{\dagger}\right)_{ik} = D^{(\alpha)}(g)_{ij}\, D^{(\beta)}(g)^{*}_{kl}.$$

Here j, l are fixed choices; the sum over a, b has collapsed. If $\alpha = \beta$, the square matrix E_{jl} has trace δ_{jl}.

Group Theory

Deriving matrix-element orthogonality

Insert $X = E_{jl}$ into the averaging result and take its (i, k) entry. We find

$$\frac{1}{N} \sum_{g} D^{(\alpha)}(g)_{ij} D^{(\beta)}(g)^{*}_{kl} = \frac{\delta_{\alpha\beta}}{d_{\alpha}} \delta_{ik} \delta_{jl}.$$

For different irreducible labels the average is zero. For the same label, the scalar average contributes δ_{jl}/d_{α} and the identity matrix contributes δ_{ik}.

Taking complex conjugates gives the usual form on the next page. The Kronecker deltas are real, so the right-hand side is unchanged.

Group Theory

The Great Orthogonality Theorem

For a chosen set of pairwise inequivalent unitary irreducible representations,

$$\boxed{\frac{1}{N}\sum_{g\in G} D^{(\alpha)}(g)^*_{ij} D^{(\beta)}(g)_{kl} = \frac{\delta_{\alpha\beta}\delta_{ik}\delta_{jl}}{d_\alpha}.}$$

The sum is over group elements. The representation labels and matrix indices on this page are fixed; none is implicitly summed.

For $\alpha = \beta$, both factors must use the same matrices in the same basis.

Zee uses $(D^\dagger)_{ij} = D^*_{ji}$ instead, so his deltas read $\delta_{il}\delta_{jk}$. The transpose exchanges the first two indices.

Textbook: Zee, II.2, pp. 103–104; Bünemann, Eqs. (5.1)–(5.2), pp. 57–58.

Group Theory

Reading the three Kronecker deltas

The right-hand side is nonzero only if

$$\alpha = \beta, \qquad i = k, \qquad j = l.$$

Thus different irreducible representations give zero, and different matrix entries of the same chosen irreducible representation give zero.

For a matching entry,

$$\frac{1}{N} \sum_{g} \left| D^{(\alpha)}(g)_{ij} \right|^2 = \frac{1}{d_\alpha}.$$

When $\alpha \neq \beta$, the dimensions can differ. The matrix-index deltas are relevant only when the irreducible labels match.

Group Theory

Orthogonality of functions on a finite group

A function $f : G \to \mathbb{C}$ is a list of N complex numbers. Define

$$\langle f, h \rangle_G = \frac{1}{N} \sum_{g \in G} f(g)^* h(g).$$

For each fixed α, i, j, the entry $D^{(\alpha)}(g)_{ij}$ is such a function. The theorem says that the functions

$$\varphi_{\alpha ij}(g) = \sqrt{d_\alpha}\, D^{(\alpha)}(g)_{ij}$$

are orthonormal. There are $\sum_\alpha d_\alpha^2$ of them in an N-dimensional space, so $\sum_\alpha d_\alpha^2 \leq N$. Lecture 10 will establish equality.

Group Theory

A numerical check for the D_3 plane matrices

Order the elements as e, r, r^2, s, rs, r^2s. The entries in the first row are

g	e	r	r^2	s	rs	r^2s
$D(g)_{11}$	1	$-1/2$	$-1/2$	1	$-1/2$	$-1/2$
$D(g)_{12}$	0	$-\sqrt{3}/2$	$\sqrt{3}/2$	0	$\sqrt{3}/2$	$-\sqrt{3}/2$

Thus $\sum_g |D(g)_{11}|^2 = 3 = N/d$, while

$$\sum_g D(g)^*_{11} D(g)_{12} = 0.$$

The theorem concerns the six values of each entry as a function of g, not just the rows of one matrix.

Group Theory

Exercise: orthogonality for $\mathbb{Z}_3$

Let $\mathbb{Z}_3 = \{e, r, r^2\}$, $\omega = e^{2\pi i/3}$, and

$$D^{(k)}(r^m) = \omega^{km}, \qquad k, m = 0, 1, 2.$$

These representations are one dimensional. Check that

$$\frac{1}{3} \sum_{m=0}^{2} D^{(k)}(r^m)^* D^{(l)}(r^m) = \delta_{kl}.$$

Use $1+\omega+\omega^2 = 0$. Which part of the Great Orthogonality Theorem has disappeared because the matrices are 1×1?

Group Theory

Exercise: solution

The product in the sum is $\omega^{(l-k)m}$. For $k = l$, all three terms are 1, so the average is 1.

If $k \neq l$, set $q = \omega^{l-k}$. Then $q \neq 1$, $q^3 = 1$, and

$$1 + q + q^2 = \frac{q^3 - 1}{q - 1} = 0.$$

Thus the average vanishes. The matrix indices have only one value, so their Kronecker deltas equal 1.

This is the finite Fourier orthogonality relation. For a cyclic group, the irreducible matrix-entry functions are the familiar complex phases.

Textbook: Zee, II.2, p. 104.

Group Theory

Lecture 9 in brief

- A matrix commuting with an irreducible complex representation is a scalar matrix.
- An intertwiner between inequivalent irreducible representations must vanish.
- Averaging $D(g)XD(g)^\dagger$ over an irreducible representation gives $(\operatorname{tr} X/d)I$.
- Choosing a matrix with one nonzero entry gives the Great Orthogonality Theorem.

Next: take traces to obtain basis-independent tools for identifying and decomposing representations.

Characters, Multiplicities and Irreducible Dimensions

Week 5 · Lecture 10

Mehmet Özkan

Group Theory

Reading for Lecture 10

- **Zee, II.2, pp. 104–109, and II.3, pp. 114–116:** character orthogonality, multiplicities, the regular representation and cyclic examples.
- **Bünemann, §4.2, pp. 52–56; §5.2, pp. 59–64:** characters, dimensions and irreducibility. For the cyclic example, see Table 5.1, p. 66.

Core: character orthogonality, multiplicities, irreducibility and the regular representation. Column orthogonality and complex characters are optional W6 previews.

We reuse the W4 S_3 matrices. W6 will recap S_3 and construct the D_4 table from scratch.

Group Theory

Characters remove basis dependence

For a representation D, define

$$\chi_D(g) = \operatorname{tr} D(g).$$

If $D'(g) = S^{-1}D(g)S$, cyclicity of the trace gives

$$\chi_{D'}(g) = \operatorname{tr}\big(S^{-1}D(g)S\big) = \operatorname{tr} D(g) = \chi_D(g).$$

Thus equivalent representations have the same character. The individual matrix entries can change under a basis transformation, while their trace remains unchanged.

We will show that, for finite groups over $\mathbb{C}$, the entire character also determines the representation up to equivalence.

Textbook: Zee, II.1, pp. 93–94; Bünemann, §4.2, p. 52.

Group Theory

A character is constant on a conjugacy class

If $g' = h^{-1}gh$, then

$$D(g') = D(h)^{-1}D(g)D(h).$$

The matrices are similar, so

$$\chi_D(g') = \chi_D(g).$$

A function constant on each conjugacy class is called a **class function**. We can specify a character by listing one value per class.

This does not say that conjugate elements have identical matrices. It says that their traces agree in every representation.

Textbook: Zee, II.1, p. 93; Bünemann, §4.2, pp. 52–53.

Group Theory

Three useful character identities

For finite-dimensional representations of our finite group,

$$\begin{aligned}\chi_D(e) &= \dim D,\\ \chi_{D\oplus D'}(g) &= \chi_D(g) + \chi_{D'}(g),\\ \chi_D(g^{-1}) &= \chi_D(g)^*.\end{aligned}$$

The first two follow from the trace of an identity or a block-diagonal matrix. For the third, choose a unitary basis and use $D(g^{-1}) = D(g)^\dagger$.

If g and g^{-1} belong to the same class, the character on that class is real. Complex characters are possible when the inverse class is different.

Group Theory

Why class sums carry weights

Let $C_1, \ldots, C_k$ be the conjugacy classes, with $n_i = |C_i|$. If f is a class function, then

$$\sum_{g \in G} f(g) = \sum_{i=1}^{k} n_i f(C_i), \qquad \sum_{i=1}^{k} n_i = N.$$

For S_3, our classes have sizes $1, 2, 3$:

$$\sum_{g \in S_3} f(g) = f(e) + 2f(r) + 3f(s).$$

The factors count how many group elements give the same value. Replacing a sum over elements by a sum over classes without these factors changes the sum.

Group Theory

Character orthogonality from matrix orthogonality

Take $i = j$ and $k = l$ in the Great Orthogonality Theorem, then sum the two diagonal indices:

$$\frac{1}{N}\sum_g \chi_\alpha(g)^* \chi_\beta(g) = \sum_{a,b} \frac{\delta_{\alpha\beta}\delta_{ab}\delta_{ab}}{d_\alpha}$$
$$= \delta_{\alpha\beta}.$$

When $\alpha = \beta$, exactly d_α pairs have $a = b$; this cancels the denominator. Consequently,

$$\boxed{\frac{1}{N}\sum_i n_i \chi_\alpha(C_i)^* \chi_\beta(C_i) = \delta_{\alpha\beta}.}$$

Textbook: Zee, II.2, p. 104; Bünemann, Eq. (5.9), pp. 59–60.

Group Theory

The inner product of class functions

For class functions f, h, the group-function inner product becomes

$$\langle f, h \rangle_G = \frac{1}{N} \sum_i n_i f(C_i)^* h(C_i).$$

The irreducible characters are orthonormal in this inner product:

$$\langle \chi_\alpha, \chi_\beta \rangle_G = \delta_{\alpha\beta}.$$

Equivalently, the vectors with components $\sqrt{n_i/N}\, \chi_\alpha(C_i)$ are orthonormal under the ordinary complex dot product. The square roots absorb the class-size weights.

Textbook: Bünemann, Eq. (5.11), p. 60.

Group Theory

The S_3 characters already obtained in W4

Write D_E for the irreducible plane representation and P for the three-dimensional permutation representation.

Class	$\{e\}$	$\{r, r^2\}$	$\{s, rs, r^2s\}$
Class size	1	2	3
χ_1	1	1	1
χ_{sgn}	1	1	-1
χ_E	2	-1	0
χ_P	3	0	1

Textbook: Bünemann, Table 4.1, p. 53. The extra row is $\chi_P = \chi_1 + \chi_E$.

Group Theory

Checking character orthogonality for S_3

For the plane representation,

$$\langle \chi_E, \chi_E \rangle_G = \frac{1}{6}(1 \cdot 2^2 + 2 \cdot (-1)^2 + 3 \cdot 0^2) = 1.$$

Its inner product with the trivial character is

$$\langle \chi_1, \chi_E \rangle_G = \frac{1}{6}(2 - 2 + 0) = 0.$$

Likewise, $\langle \chi_1, \chi_{\text{sgn}} \rangle_G = (1 + 2 - 3)/6 = 0$, and $\langle \chi_{\text{sgn}}, \chi_E \rangle_G = (2 - 2 + 0)/6 = 0$.
The weights are $1, 2, 3$, not equal weights for the three columns.

Group Theory

How many independent class functions are there?

A class function is determined by its k values on the k conjugacy classes. Thus the space of class functions has dimension k.

For each class, define an indicator function

$$\mathbf{1}_{C_i}(g) = \begin{cases} 1, & g \in C_i, \\ 0, & g \notin C_i. \end{cases}$$

These k functions are independent and span the space. Since distinct irreducible characters are orthonormal class functions, the number of inequivalent irreducible representations is at most k.

We will prove the reverse inequality after studying the regular representation.

Group Theory

Multiplicity and dimension are different numbers

Complete reducibility gives

$$D \simeq \bigoplus_{\alpha} \left(D^{(\alpha)}\right)^{\oplus m_\alpha}, \qquad m_\alpha = 0, 1, 2, \ldots .$$

The integer m_α counts copies of one irreducible representation. Taking traces and dimensions gives

$$\chi_D = \sum_{\alpha} m_\alpha \chi_\alpha, \qquad \dim D = \sum_{\alpha} m_\alpha d_\alpha .$$

Keep the symbols separate: N is the group order, n_i is a class size, d_α is an irreducible dimension, and m_α is a multiplicity in the chosen D.

Textbook: Zee, II.2, pp. 106–107; Bünemann, Eq. (4.16), p. 53.

Group Theory

Deriving the multiplicity formula

Take the inner product of $\chi_D = \sum_\beta m_\beta \chi_\beta$ with one irreducible character:

$$\langle \chi_\alpha, \chi_D \rangle_G = \sum_\beta m_\beta \langle \chi_\alpha, \chi_\beta \rangle_G = m_\alpha.$$

Hence

$$\boxed{m_\alpha = \frac{1}{N} \sum_i n_i \chi_\alpha(C_i)^* \chi_D(C_i).}$$

For the character of an actual representation, the answer is a nonnegative integer. We can find all multiplicities without constructing the matrix that makes D block diagonal.

Textbook: Zee, II.2, Eq. (9), p. 107; Bünemann, Eq. (5.23), p. 63.

Group Theory

Example: decomposing the permutation representation

For P, the character values are $(3, 0, 1)$ on classes of sizes $(1, 2, 3)$. Thus

$$m_1 = \frac{1}{6}(3 + 0 + 3) = 1,$$
$$m_{\mathrm{sgn}} = \frac{1}{6}(3 + 0 - 3) = 0,$$
$$m_E = \frac{1}{6}(2 \cdot 3 + 0 + 0) = 1.$$

Therefore $P \simeq 1 \oplus D_E$, agreeing with our explicit W4 basis change. The dimension check is $3 = 1 \cdot 1 + 1 \cdot 2$.

Group Theory

A character test for irreducibility

Orthogonality gives the squared norm of the character:

$$\langle \chi_D, \chi_D \rangle_G = \sum_{\alpha,\beta} m_\alpha m_\beta \delta_{\alpha\beta} = \sum_\alpha m_\alpha^2.$$

For a nonzero representation, the multiplicities are nonnegative integers. Therefore

$$\boxed{D \text{ irreducible} \quad \Longleftrightarrow \quad \frac{1}{N} \sum_i n_i |\chi_D(C_i)|^2 = 1.}$$

An answer greater than 1 means reducibility. The test applies to characters of representations; a general class function with norm one need not be an irreducible character.

Textbook: Zee, II.2, pp. 106–107; Bünemann, §5.2.3, p. 64.

Group Theory

What the squared character norm counts

The squared norm counts $\sum_\alpha m_\alpha^2$, rather than the dimension or the total number of blocks.

D	Dimension	Copies in total	$\langle \chi_D, \chi_D \rangle_G$
D_E	2	1	1
$1 \oplus D_E$	3	2	2
$1 \oplus 1$	2	2	4
$1 \oplus 1 \oplus D_E$	4	3	5

For two identical trivial blocks, $m_1 = 2$ and the squared norm is $2^2 = 4$.

Group Theory

Exercise: a four-dimensional representation

Suppose an S_3 representation has character values

$$\chi_D = (4, 1, 0)$$

on our three classes, of sizes $1, 2, 3$.

1. Calculate its squared norm. Is it irreducible?
2. Calculate m_1, m_{sgn}, m_E using the W4 characters.
3. Check both the dimension and the character on every class against the resulting decomposition.

Group Theory

Exercise: solution

The squared norm is $(16+2)/6 = 3$, so the representation is reducible. Its multiplicities are

$$m_1 = \frac{4+2}{6} = 1, \qquad m_{\mathrm{sgn}} = \frac{4+2}{6} = 1, \qquad m_E = \frac{8-2}{6} = 1.$$

Thus

$$D \simeq 1 \oplus D_{\mathrm{sgn}} \oplus D_E, \qquad 4 = 1+1+2.$$

Adding the three irreducible characters gives $(1,1,1) + (1,1,-1) + (2,-1,0) = (4,1,0)$. The squared norm agrees with $1^2 + 1^2 + 1^2 = 3$.

Group Theory

Why the character determines the representation

Suppose D and D' have the same character on every class. Their multiplicity formulas then give

$$m_\alpha(D) = m_\alpha(D') \qquad \text{for every } \alpha.$$

Both representations are direct sums of the same irreducible representatives with the same multiplicities. They are therefore equivalent.

Conversely, equivalent representations have equal traces. Hence, for finite groups over $\mathbb{C}$,

$$\boxed{D \simeq D' \quad \Longleftrightarrow \quad \chi_D = \chi_{D'}.}$$

This concerns the full character, not its value on just one class.

Textbook: Zee, II.1, p. 94, and II.2, p. 107; Bünemann, p. 63.

Group Theory

The regular representation

Give a complex vector space one independent basis vector e_g for every $g \in G$. It has dimension N. Define

$$L(h)e_g = e_{hg}.$$

Then $L(h)L(k)e_g = e_{hkg} = L(hk)e_g$, so L is a representation. Each $L(h)$ is a permutation matrix.

For S_3, this is a six-dimensional representation. The three-dimensional permutation representation of W4 instead had one basis vector per vertex. Both are permutation representations, but they act on different sets.

Textbook: Zee, II.2, pp. 107–108; Bünemann, §4.2, pp. 53–54.

Group Theory

The regular character

A diagonal entry of $L(h)$ equals 1 precisely when $L(h)e_g = e_g$, that is, when $hg = g$. Multiplication by g^{-1} gives $h = e$. Thus $L(e) = I_N$ fixes every label, and $L(h)$ fixes no labels when $h \neq e$:

$$\boxed{\chi_{\mathrm{reg}}(h) = \begin{cases} N, & h = e, \\ 0, & h \neq e. \end{cases}}$$

This character is easy to compute even though the regular matrices may be large.

Textbook: Zee, II.2, p. 108; Bünemann, §4.2, p. 54.

Group Theory

Every irreducible representation appears in the regular one

For any irreducible character χ_α, the multiplicity formula gives

$$m_\alpha^{\mathrm{reg}} = \frac{1}{N}\sum_g \chi_\alpha(g)^* \chi_{\mathrm{reg}}(g) = \chi_\alpha(e)^* = d_\alpha.$$

Only the identity term contributes. Therefore

$$\boxed{L \simeq \bigoplus_\alpha \left(D^{(\alpha)}\right)^{\oplus d_\alpha}.}$$

In particular, no irreducible representation is absent. A two-dimensional irreducible appears twice, contributing four dimensions to the regular space.

Textbook: Zee, II.2, p. 108; Bünemann, Eq. (5.16), p. 61.

Group Theory

The sum of squared irreducible dimensions

The regular space has dimension N. Its decomposition therefore gives

$$N = \sum_\alpha m_\alpha^{\mathrm{reg}} d_\alpha = \sum_\alpha d_\alpha^2.$$

Thus

$$\boxed{\sum_\alpha d_\alpha^2 = N.}$$

The sum contains one term for each inequivalent irreducible representation. It does not sum the squares of all possible reducible representations.

Together with the number of conjugacy classes, this strongly constrains the possible irreducible dimensions.

Textbook: Zee, II.2, p. 108; Bünemann, Eq. (4.20), p. 54.

Group Theory

The regular representation of S_3

The irreducible representations we already know have dimensions $1, 1, 2$. Since

$$1^2 + 1^2 + 2^2 = 6 = |S_3|,$$

they exhaust the dimension sum. There can be no further inequivalent irreducible representation.

The regular decomposition is

$$L \simeq 1 \oplus D_{\mathrm{sgn}} \oplus D_E \oplus D_E.$$

Its character is $(1, 1, 1) + (1, 1, -1) + 2(2, -1, 0) = (6, 0, 0)$. The dimension is $1 + 1 + 2 + 2 = 6$, and the squared norm is $1 + 1 + 4 = 6$.

Group Theory

Why Abelian groups have one-dimensional complex irreps

If G is Abelian, every $D(g)$ commutes with every $D(h)$. In an irreducible complex representation, Schur's lemma therefore gives

$$D(g) = \lambda_g I \qquad \text{for every } g \in G.$$

If the space had dimension greater than one, every line would then be invariant. Irreducibility consequently forces $d = 1$.

The dimension sum now shows that a finite Abelian group of order N has exactly N inequivalent complex irreducible representations. Each contributes 1^2 to the sum.

Textbook: Zee, II.3, p. 116; Bünemann, p. 56.

Group Theory

The number of irreps equals the number of classes

Let r be the number of inequivalent complex irreducible representations and k the number of conjugacy classes. The theorem is

$$\boxed{r = k.}$$

We already proved $r \leq k$: irreducible characters are orthonormal vectors in the k-dimensional space of class functions.

For completeness, the next three pages prove $k \leq r$ using class sums and the regular representation. This proof can be read separately without interrupting the use of the character formulas.

Textbook: Zee, II.2, pp. 108–109 and appendix 4; Bünemann, pp. 61–63.

Group Theory

Optional proof: class sums act as scalars

For each class C_i and irreducible representation $D^{(\alpha)}$, form

$$K_i^{(\alpha)} = \sum_{g \in C_i} D^{(\alpha)}(g).$$

Conjugation by $D^{(\alpha)}(h)$ merely permutes the elements of C_i. Hence $K_i^{(\alpha)}$ commutes with the representation and is scalar by Schur's lemma.

Taking its trace determines that scalar:

$$K_i^{(\alpha)} = \lambda_{\alpha i} I_{d_\alpha}, \qquad \lambda_{\alpha i} = \frac{n_i \chi_\alpha(C_i)}{d_\alpha}.$$

Textbook: Bünemann, Eqs. (5.18)–(5.20), pp. 61–62.

Group Theory

Optional proof: independent regular class sums

In the regular representation, set $K_i = \sum_{g \in C_i} L(g)$. These k matrices are linearly independent.

Indeed, applying a possible relation to the basis vector e_e gives

$$\left(\sum_i t_i K_i \right) e_e = \sum_i t_i \sum_{g \in C_i} e_g = 0.$$

The classes are disjoint, and the e_g form a basis. Therefore every coefficient t_i must vanish.

Thus the vector space spanned by the regular class-sum matrices has dimension k.

Group Theory

Optional proof: at most one scalar per irrep

In the irreducible decomposition of the regular representation, K_i acts by $\lambda_{\alpha i} I$ on every copy of $D^{(\alpha)}$. Associate to it the vector

$$K_i \longmapsto (\lambda_{1i}, \ldots, \lambda_{ri}) \in \mathbb{C}^r.$$

If a linear combination has the zero vector of scalars, it acts as zero on every irreducible block, and hence is the zero regular matrix. This linear map is injective.

It maps a k-dimensional space into $\mathbb{C}^r$, so $k \leq r$. Together with $r \leq k$, this proves $r = k$. The irreducible characters therefore form a basis of all class functions.

Group Theory

W6 preview: rows and columns

We place irreps in rows and classes in columns, as in Bünemann and W4. Zee uses the transposed arrangement.

In our convention, row orthogonality is

$$\frac{1}{N}\sum_i n_i \chi_\alpha(C_i)^* \chi_\beta(C_i) = \delta_{\alpha\beta}.$$

Column orthogonality is

$$\sum_\alpha \chi_\alpha(C_i)^* \chi_\alpha(C_j) = \frac{N}{n_i}\delta_{ij}.$$

Textbook: Zee, II.2, pp. 104–105, 109; Bünemann, p. 59.

Group Theory

W6 preview: column orthogonality

Normalize the character table as

$$U_{\alpha i} = \sqrt{\frac{n_i}{N}}\,\chi_\alpha(C_i).$$

Row orthogonality gives $UU^\dagger = I$. Using $r = k$, whose proof is on the optional class-sum pages, U is square. Thus $U^\dagger U = I$, and its (i, j) entry gives

$$\frac{\sqrt{n_i n_j}}{N} \sum_\alpha \chi_\alpha(C_i)^* \chi_\alpha(C_j) = \delta_{ij}.$$

For $i = j$, the prefactor is n_i/N. For different columns, the sum vanishes. This is column orthogonality in our table convention.

Group Theory

W6 preview: column check for S_3

Sum over the three irreducible characters. The squared lengths of the columns are

$$\begin{aligned} \{e\}: &\quad 1^2 + 1^2 + 2^2 = 6 = N/1, \\ \{r, r^2\}: &\quad 1^2 + 1^2 + (-1)^2 = 3 = N/2, \\ \{s, rs, r^2 s\}: &\quad 1^2 + (-1)^2 + 0^2 = 2 = N/3. \end{aligned}$$

For the identity and rotation columns, the cross sum is

$$1 \cdot 1 + 1 \cdot 1 + 2 \cdot (-1) = 0.$$

Unlike row orthogonality, column orthogonality sums over irreducible representations, not over group elements or classes.

Group Theory

W6 preview: complex characters of $\mathbb{Z}_3$

For $\omega = e^{2\pi i/3}$, the three irreducible characters are

	e	r	r^2
χ_0	1	1	1
χ_1	1	ω	ω^2
χ_2	1	ω^2	ω

Every class has size one. The inverse elements r, r^2 belong to different classes, and their character values are complex conjugates.

This is also the A_3 example at the beginning of Zee II.3, since $A_3 = \langle (123) \rangle \cong \mathbb{Z}_3$.

Textbook: Zee, II.3, pp. 115–116; Bünemann, Table 5.1, p. 66.

Group Theory

W6 preview: why conjugation matters

For the $\mathbb{Z}_3$ character $\chi_1 = (1, \omega, \omega^2)$,

$$\langle \chi_1, \chi_1 \rangle_G = \frac{1}{3}(1 + |\omega|^2 + |\omega^2|^2) = 1.$$

Without complex conjugation we would instead obtain

$$\frac{1}{3}(1 + \omega^2 + \omega^4) = \frac{1}{3}(1 + \omega^2 + \omega) = 0.$$

This is not the character inner product. The conjugate on the first character is needed even when many familiar examples happen to have real entries.

Group Theory

Optional application: projecting onto invariant vectors

For any unitary representation, define

$$\Pi_0 = \frac{1}{N} \sum_g D(g).$$

For every h, $D(h)\Pi_0 = \Pi_0$ by relabelling, so the image consists of fixed vectors. On fixed vectors, Π_0 acts as the identity. Applying it twice therefore gives $\Pi_0^2 = \Pi_0$.

Replacing g with g^{-1} gives $\Pi_0^\dagger = \Pi_0$. Together these properties make Π_0 the orthogonal projector onto the fixed vectors. Its trace is

$$\operatorname{tr} \Pi_0 = \frac{1}{N} \sum_g \chi_D(g) = m_1,$$

the multiplicity of the trivial representation.

Group Theory

Example: the invariant projector for three vertices

For the three-dimensional permutation representation, averaging all six permutation matrices gives

$$\Pi_0 = \frac{1}{3} J_3, \qquad \Pi_0 \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{x+y+z}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

This is precisely the constant-vector component found in W4. The remaining projector $I - \Pi_0$ selects the zero-sum plane.

The character calculation gives the rank directly:

$$\operatorname{tr} \Pi_0 = \frac{1}{6}(3 + 2 \cdot 0 + 3 \cdot 1) = 1.$$

Thus the permutation representation contains exactly one trivial copy.

Group Theory

Exercise: repeated irreducible pieces

An S_3 representation has character

$$\chi_D = (7, -2, 1)$$

on classes of sizes $1, 2, 3$.

1. Find the multiplicities of 1, D_{sgn} and D_E.
2. Calculate $\dim D$, the total number of irreducible copies and the squared character norm. Why are these answers different?
3. Find the dimension of the space of vectors fixed by every group element.

Group Theory

Exercise: solution

The multiplicities are

$$m_1 = \frac{7-4+3}{6} = 1, \quad m_{\mathrm{sgn}} = \frac{7-4-3}{6} = 0, \quad m_E = \frac{14+4}{6} = 3.$$

Hence $D \simeq 1 \oplus 3D_E$. Here $3D_E$ means three direct-sum copies. The dimension is $1 + 3 \cdot 2 = 7$, while the number of copies is $1 + 3 = 4$.

The squared character norm is

$$\frac{1}{6}(49 + 8 + 3) = 10 = 1^2 + 3^2.$$

The fixed-vector space has dimension $m_1 = 1$. It comes from the trivial copy; none of the three plane copies has a nonzero fixed vector.

Group Theory

Checks to make when using the formulas

- Include the class sizes when replacing a group sum by a class sum. Their total must equal N.
- Conjugate the first character in an inner product. The identity character value must equal the dimension.
- For an actual representation, multiplicities must be nonnegative integers and satisfy $\dim D = \sum_\alpha m_\alpha d_\alpha$.
- Check the assumptions: the scalar form of Schur's lemma uses complex irreducibility; the conjugated matrix-entry formula uses unitary representatives.

Faithfulness remains a separate question about the representation kernel. It does not follow from irreducibility.

Group Theory

Formula recap: Schur and matrix orthogonality

For an irreducible complex representation,

$$AD(g) = D(g)A \ \ (\forall g) \quad \Longrightarrow \quad A = \lambda I.$$

Between inequivalent irreducible representations, every intertwiner is zero. Averaging over one unitary irreducible representation gives

$$\frac{1}{N}\sum_g D(g) X D(g)^\dagger = \frac{\operatorname{tr} X}{d} I_d.$$

For a fixed set of unitary irreducible representatives,

$$\frac{1}{N}\sum_g D^{(\alpha)}(g)^*_{ij} D^{(\beta)}(g)_{kl} = \frac{\delta_{\alpha\beta}\delta_{ik}\delta_{jl}}{d_\alpha}.$$

Group Theory

Formula recap: characters and multiplicities

For classes C_i of size n_i,

$$\langle f, h\rangle_G = \frac{1}{N}\sum_i n_i f(C_i)^* h(C_i), \qquad \langle \chi_\alpha, \chi_\beta\rangle_G = \delta_{\alpha\beta}.$$

For a nonzero representation D,

$$m_\alpha = \langle \chi_\alpha, \chi_D\rangle_G, \qquad D \text{ irreducible } \iff \langle \chi_D, \chi_D\rangle_G = 1.$$

The complete set of inequivalent complex irreducibles satisfies

$$\sum_\alpha d_\alpha^2 = N, \qquad \#\{\text{irreducible labels}\} = \#\{\text{conjugacy classes}\}.$$

Group Theory

Week 5 in brief

- Schur's lemma turns symmetry-commuting maps into scalars on irreducible complex spaces.
- Averaging turns that statement into orthogonality of matrix entries and characters.
- Character inner products determine irreducible multiplicities and test irreducibility.
- The regular representation gives $m_\alpha^{\mathrm{reg}} = d_\alpha$ and $\sum_\alpha d_\alpha^2 = N$. There are as many irreducible labels as classes.

Next week: recap S_3, construct the D_4 character table from scratch, and apply character decompositions to concrete representations.